

Ternary / Conditional operator(? :)

It is similar to if else / ladder if in working style.

It allows to complete if else / ladder if in a single statement.

When we are working with if else/ladder if it is going to take more than one line of statements. Ternary operator is going to finish the same task in a single statement.

But the difference between if ...else and ternary operator is ternary operator supports only one statement at a time and if supports any number of statements.

It is having 3 expressions. Hence it is called ternary operator.

It is starting with a condition. Hence it is called conditional operator.

Syntax:

condition ? true statement : false statement ;
 ↑ ↑ ↑
exp1/op1 exp2/op2 exp3/op3

If condition true, statement after ? executed.

If condition false, statement after : is executed.

When compared with if else, conditional operator **performance is high**.

Find max in 2 no's using single statement/Ternary operator:

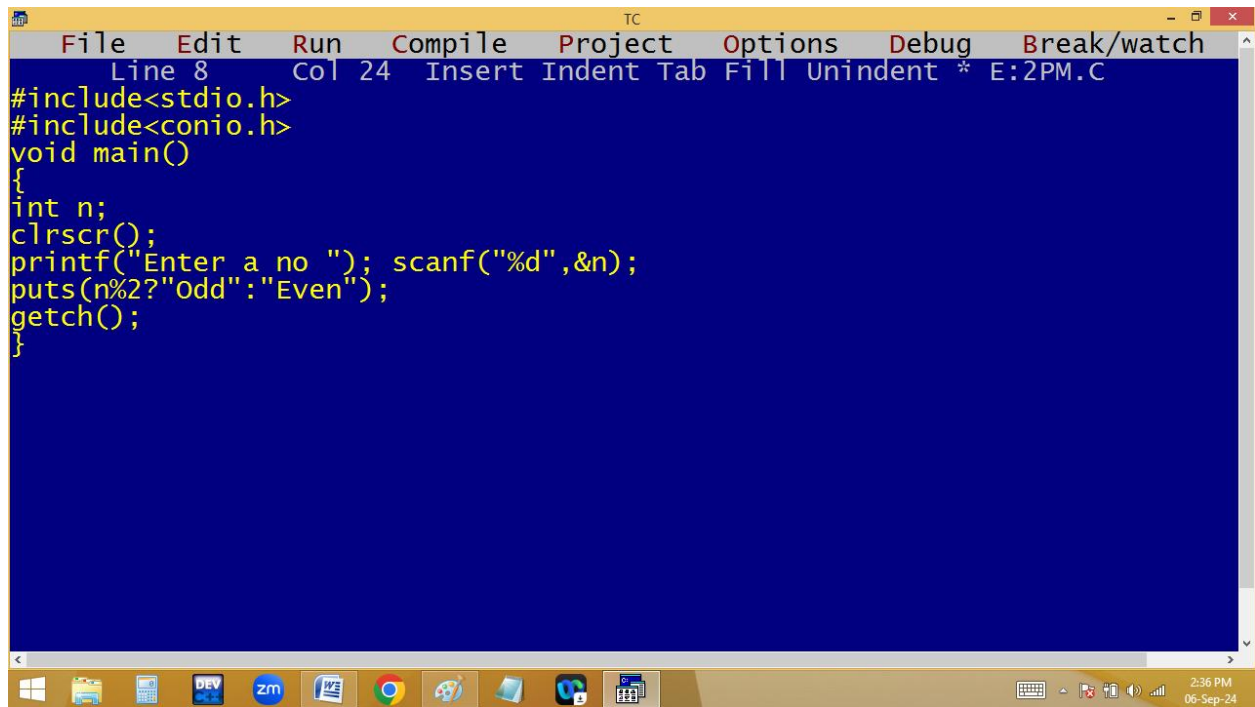
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 20 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,big;
clrscr();
printf("Enter a, b values "); scanf("%d %d",&a,&b);
puts(a>b?"a is big":"b is big");
printf("%d is big\n",a>b?a:b);
a>b?puts("a is big"):puts("b is big");
big=a>b?a:b;
printf("%d is big\n",big);
puts(a>b?"a is big": b>a ?"b is big":"both are equal");
getch();
}
```

```
TC
Enter a, b values 5 5
b is big
5 is big
b is big
5 is big
both are equal
```

```
TC
Enter a, b values 1 2
b is big
2 is big
b is big
2 is big
b is big
```

```
TC
Enter a, b values 4 3
a is big
4 is big
a is big
4 is big
a is big
```

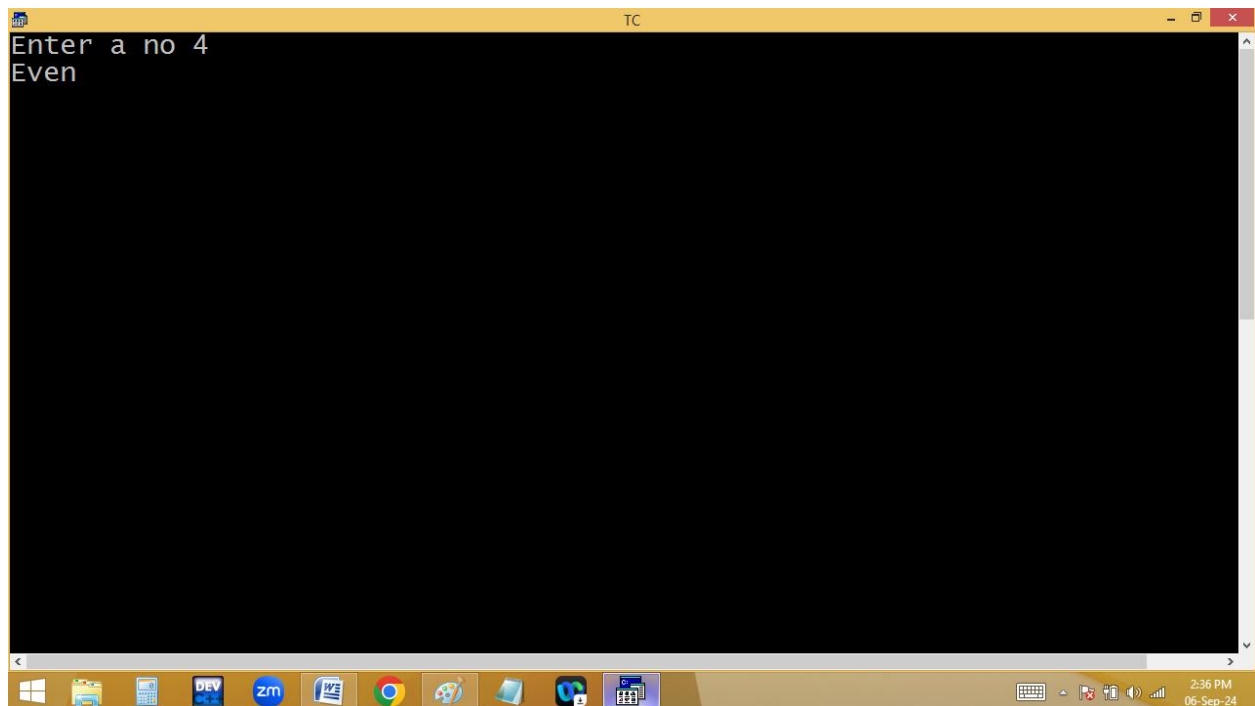
Finding even/odd using ternary operators?



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and a menu bar. The menu items are File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n;
    clrscr();
    printf("Enter a no "); scanf("%d",&n);
    puts(n%2?"Odd":"Even");
    getch();
}
```

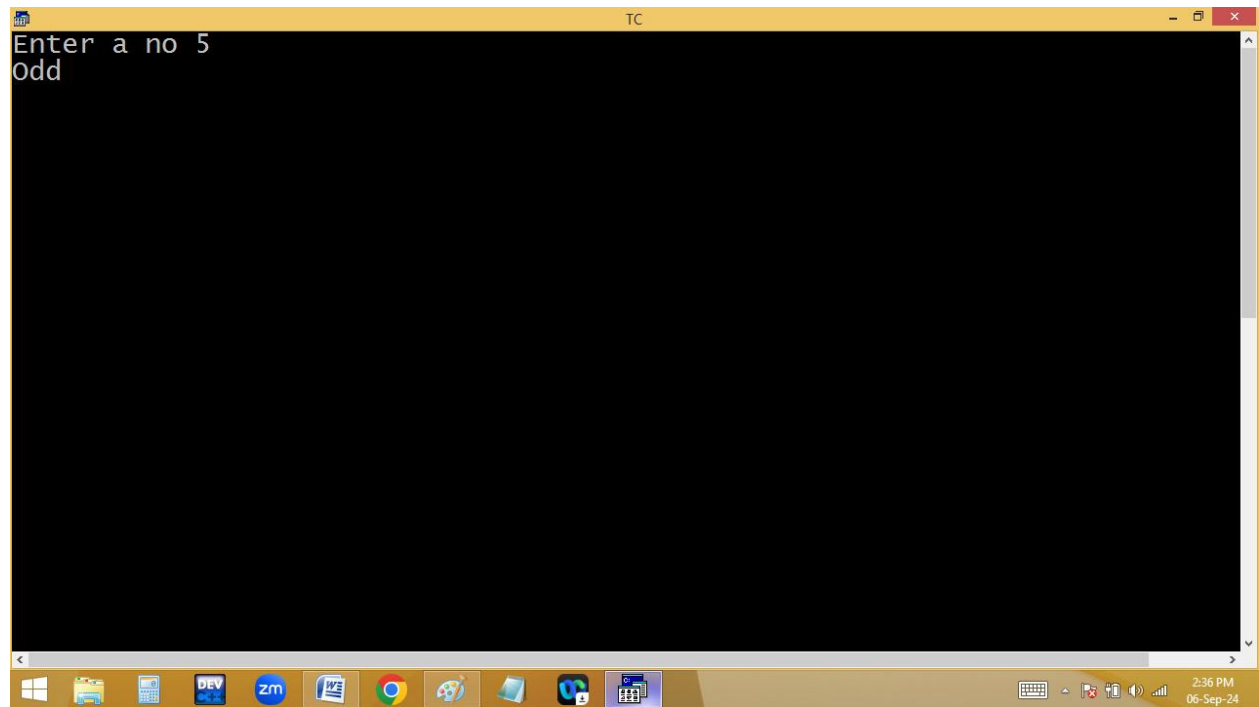
The Windows taskbar is visible at the bottom with various application icons and a system clock showing 2:36 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area now has a black background and displays the output of the program:

```
Enter a no 4
Even
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock.

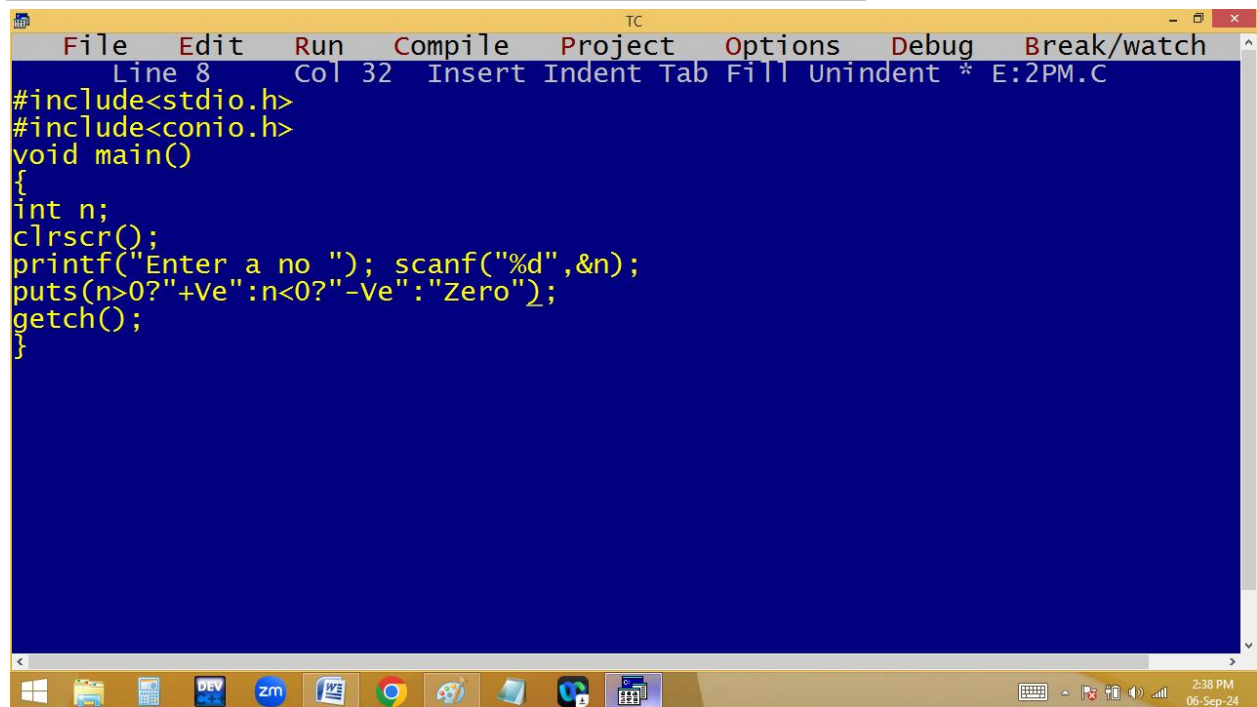


The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program in a blue editor window. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 24 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
puts(n%2==0?"Even":"Odd");
getch();
}
```

The bottom screenshot shows the same IDE with the program executed. The output window is black, and the text "Enter a no 5" and "Odd" is visible on the first two lines, indicating that the user entered the number 5 and the program correctly identified it as odd.

Finding +Ve/-Ve/0 using ternary operator?

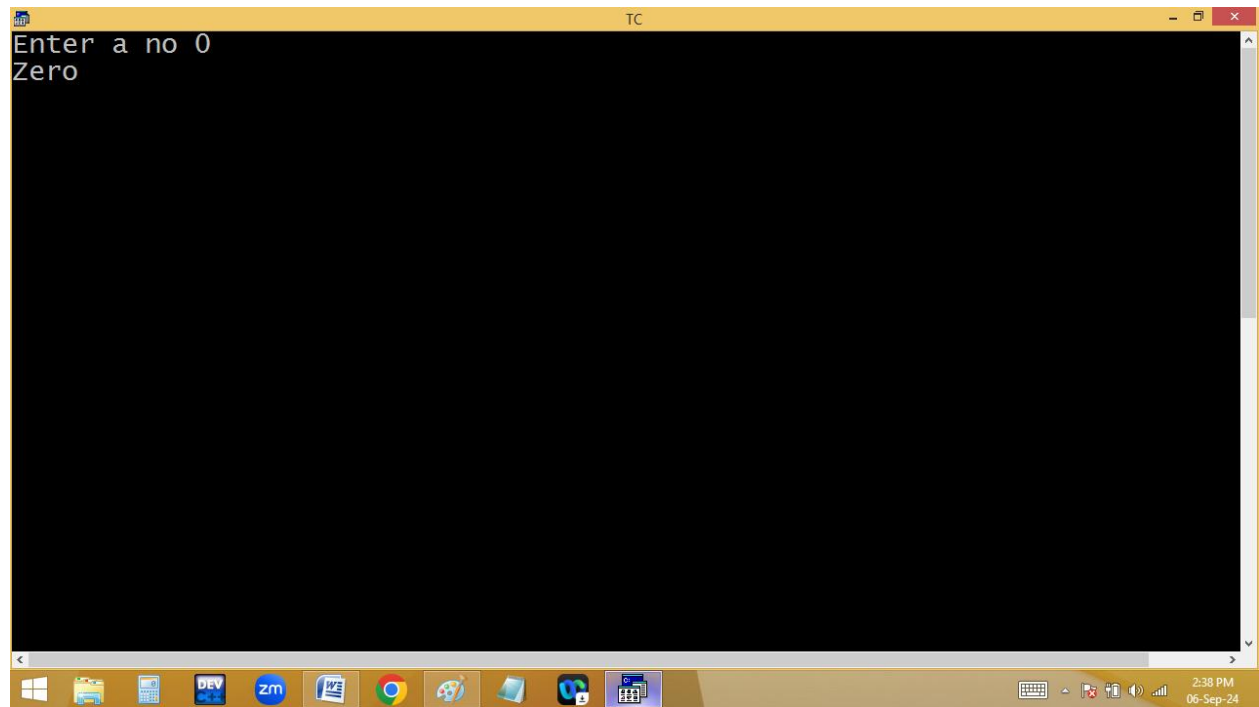


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 32 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
puts(n>0?" +Ve":n<0?" -Ve":"Zero");
getch();
}
```

The screenshot shows a Turbo C++ IDE window titled 'TC'. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 8 Col 32 Insert Indent Tab Fill Unindent * E:2PM.C'. The main editing area has a blue background and contains the following C code: `#include<stdio.h>`, `#include<conio.h>`, `void main()`, `{`, `int n;`, `clrscr();`, `printf("Enter a no "); scanf("%d",&n);`, `puts(n>0?" +Ve":n<0?" -Ve":"Zero");`, `getch();`, and `}`. The Windows taskbar is visible at the bottom with various application icons and a system clock showing 2:38 PM on 06-Sep-24.

```
TC
Enter a no 3
+Ve
```

```
TC
Enter a no -2
-Ve
```



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",0?1?2:3:4);
printf("%d\n",1?0?2:3:4);
printf("%d\n",1?2?0:3:4);
getch();
}

4
3
0
```

con ? t : f
 0 ? 1 ? 2 : 3 : 4 ✓
 con ? true : false

false {
 if (0)
 {
 if (1) p(2);
 else p(3);
 }
 else p(4);
 }

con ? t : f
 1 ? 0 ? 2 : 3 : 4
 con ? true : false

if (1)
 {
 ↓ T
 if (0) p(2);
 else p(3);
 }
 else p(4);

con ? t : f
 1 ? 2 ? 0 : 3 : 4
 con ? true : false

if (1)
 {
 ↓ T
 if (2) p(0);
 else p(3);
 }
 else p(4);

Switch statement:

It is a selection statement.

It is used to execute one case of statements from no of cases according to the switch expression value matched with case expression value. In switch the program is jumped to matching case like the go to label.

It is similar to ladder if in working style.

Switch performance is high when compared with ladder if because of it jumps to matching case.

Syntax:

```
switch(condition / expression)
```

```
{  
case  constexp1:  
statements;  
break;  
case  constexp2:  
statements;  
break;  
case  constexpN:  
statements;  
break;  
[ default: statements; ]  
}
```

Here switch, case, break, default are the keywords.

In between case and case expression / value at least one space should be provided. Otherwise it will become a label.

case expression/value should be a constant integer/char value. i.e. float / string not allowed.

One case contains one expression only.

case expression doesn't contain any separators like , . etc.

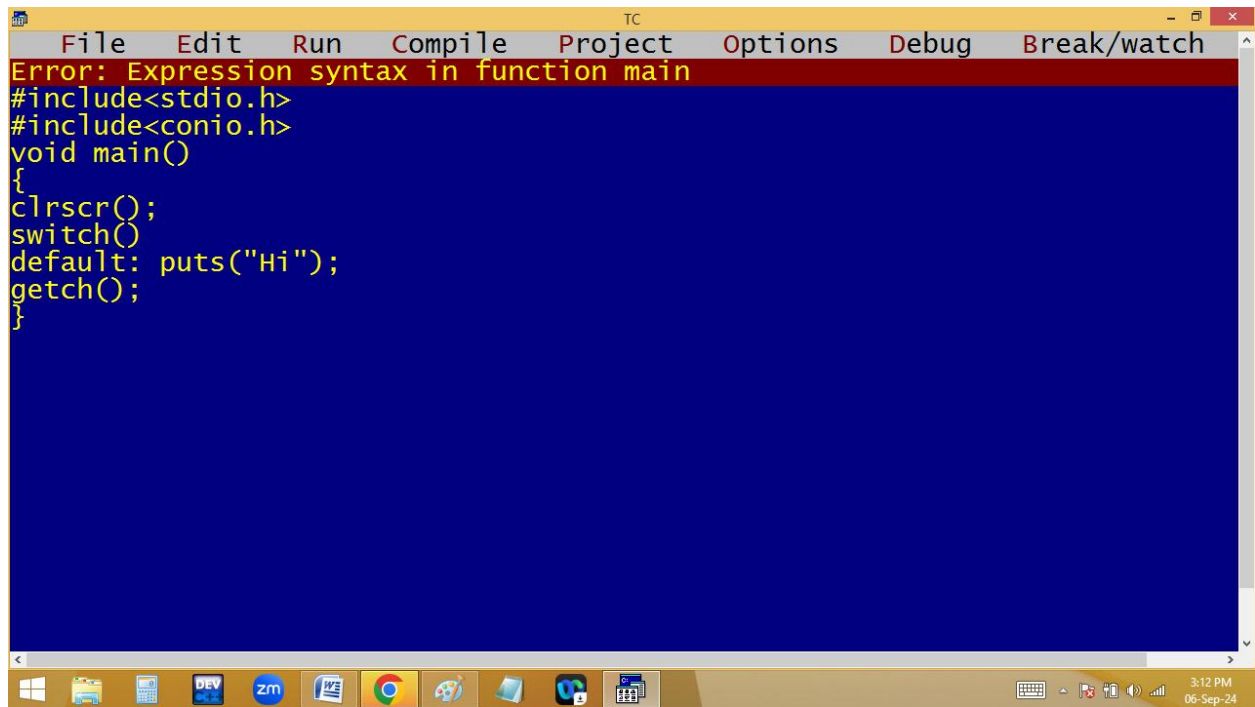
case expression should be end with : (colon)

Each case should be separated with break keyword. Otherwise remaining cases also executed.

Duplicate cases not allowed.

default is similar to the else and all cases are failed then default statements are executed. Default is optional and we can declare it anywhere in our switch.

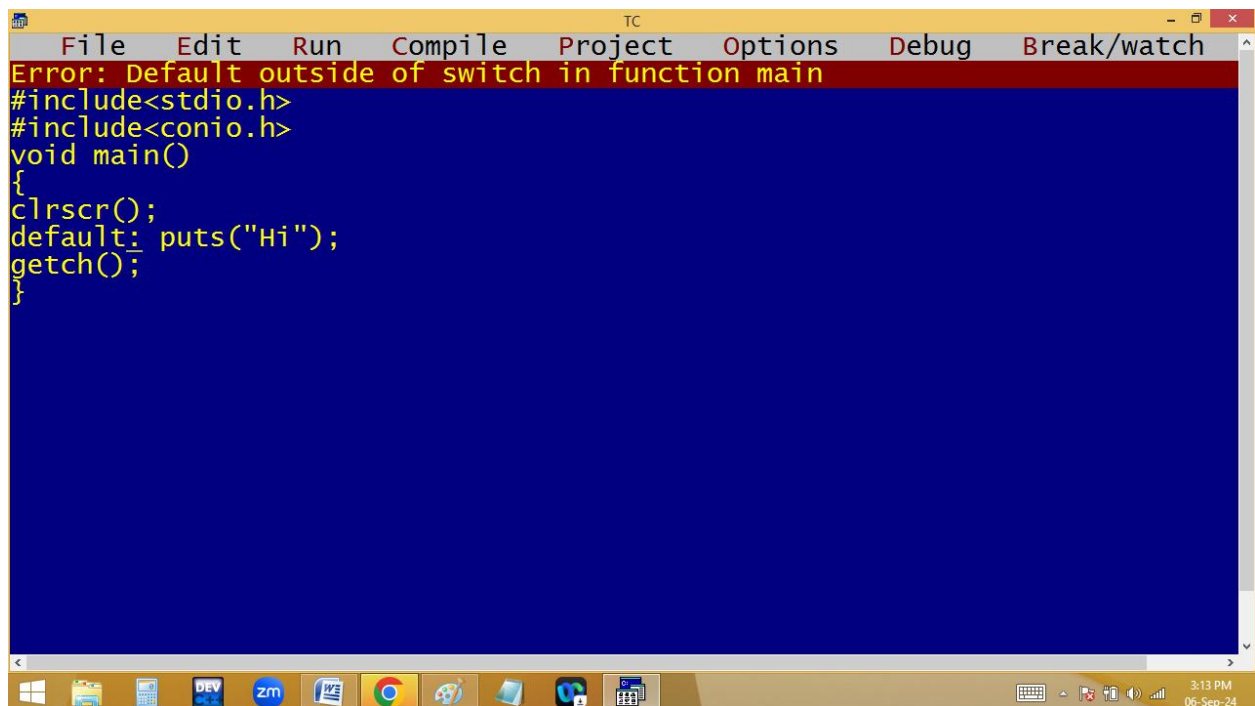
Outside case expressions not considered in switch.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a blue editor area. A red error message bar at the top reads "Error: Expression syntax in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch()
default: puts("Hi");
getch();
}
```

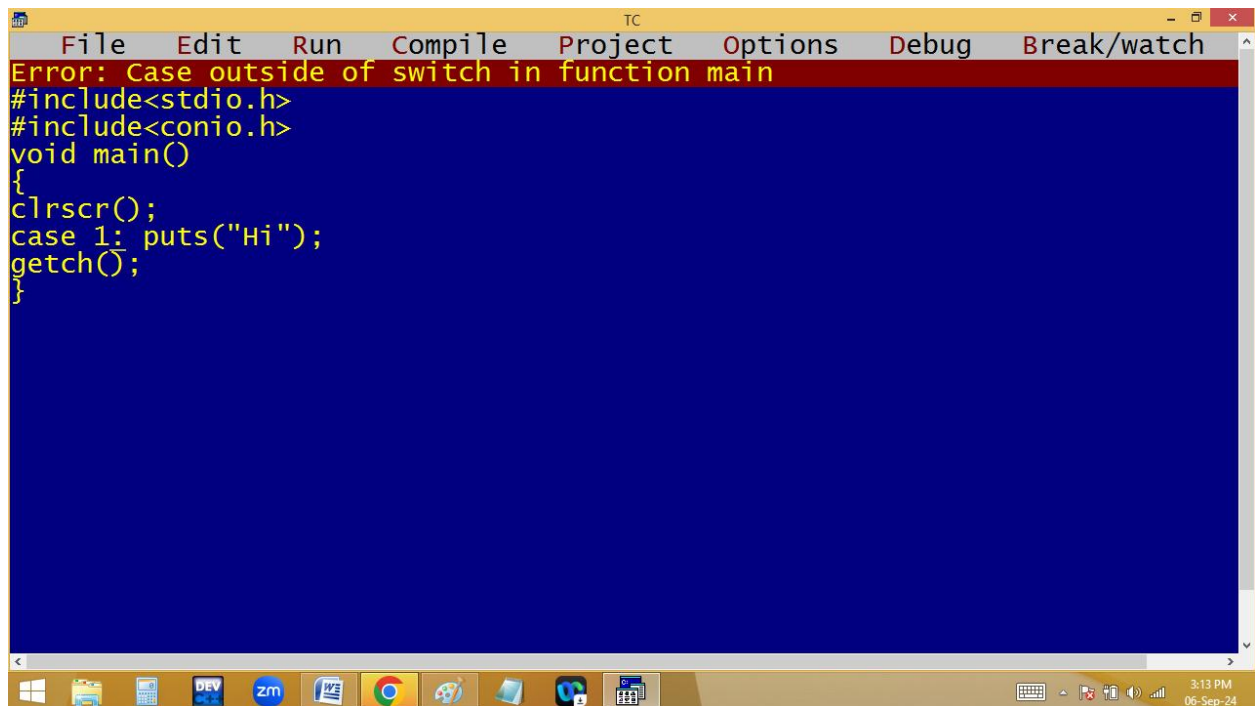
The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:12 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar. A red error message bar at the top reads "Error: Default outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
default: puts("Hi");
getch();
}
```

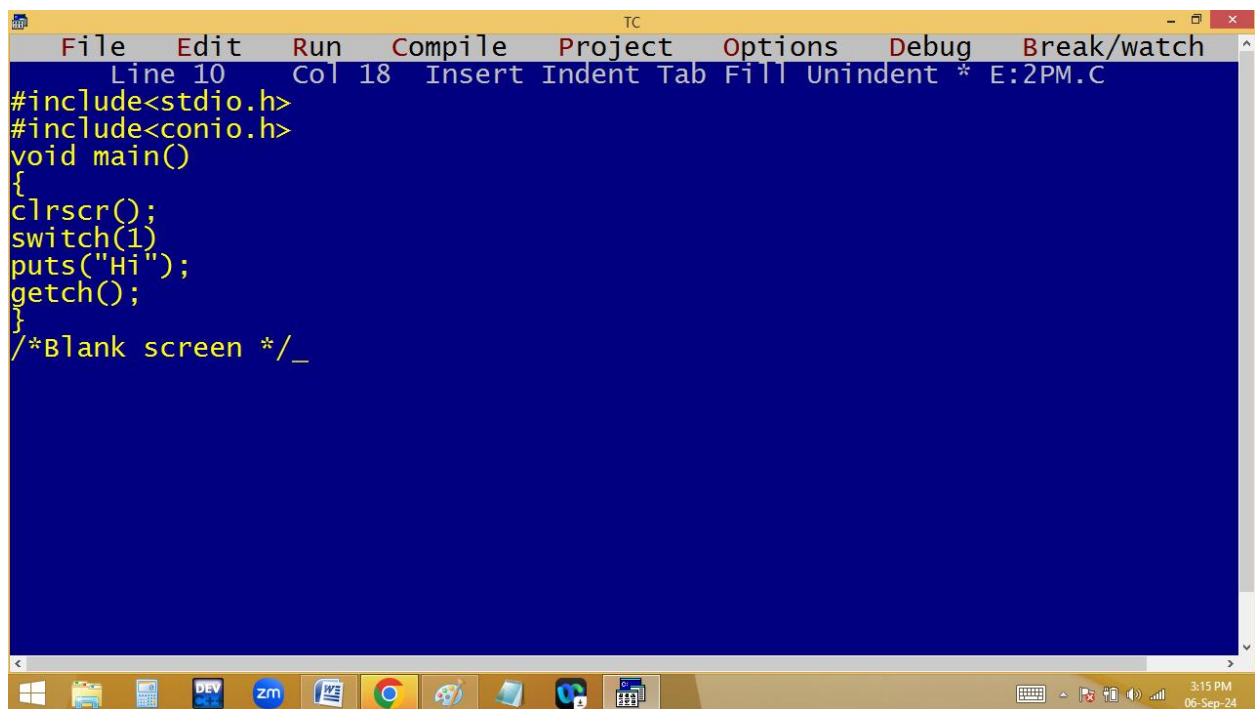
The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:13 PM on 06-Sep-24.



The screenshot shows the Turbo C++ IDE with a red error message at the top: "Error: Case outside of switch in function main". The code in the editor is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Error: Case outside of switch in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
case 1: puts("Hi");
getch();
}
```

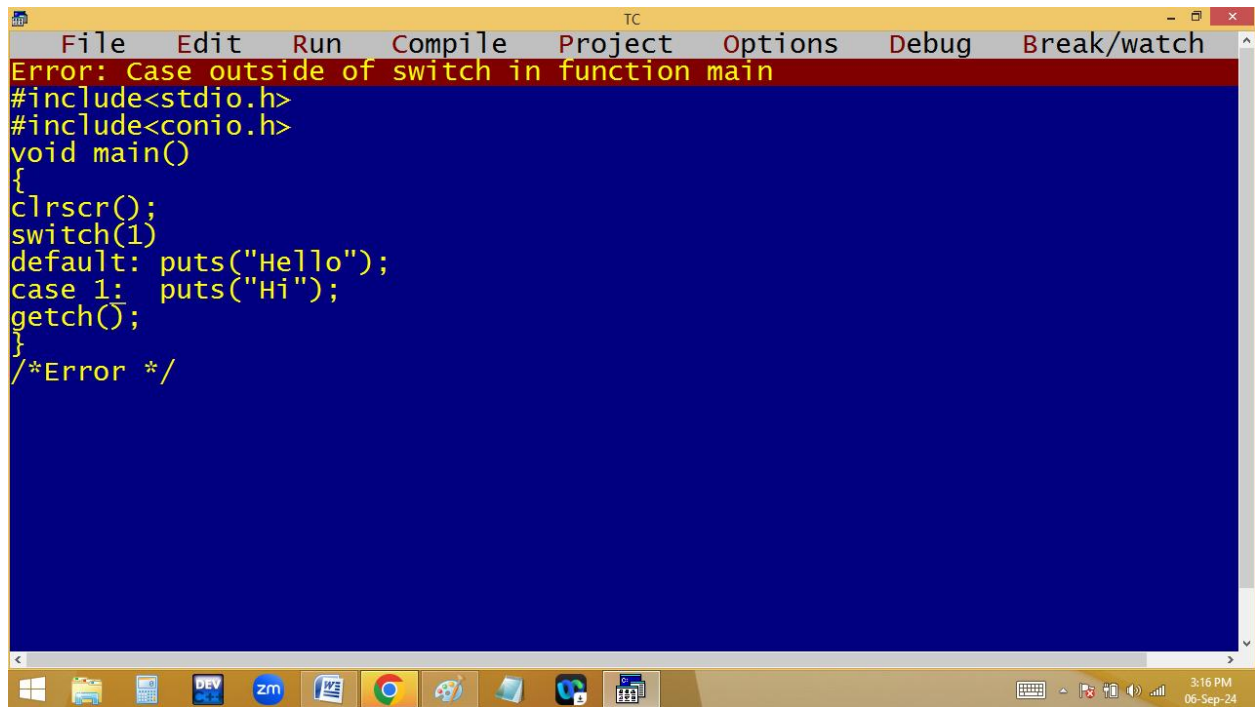
The Windows taskbar at the bottom shows the time as 3:13 PM on 06-Sep-24.



The screenshot shows the Turbo C++ IDE with the error resolved. The status bar at the top indicates "Line 10 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C". The code in the editor is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
puts("Hi");
getch();
}
/*Blank screen */_
```

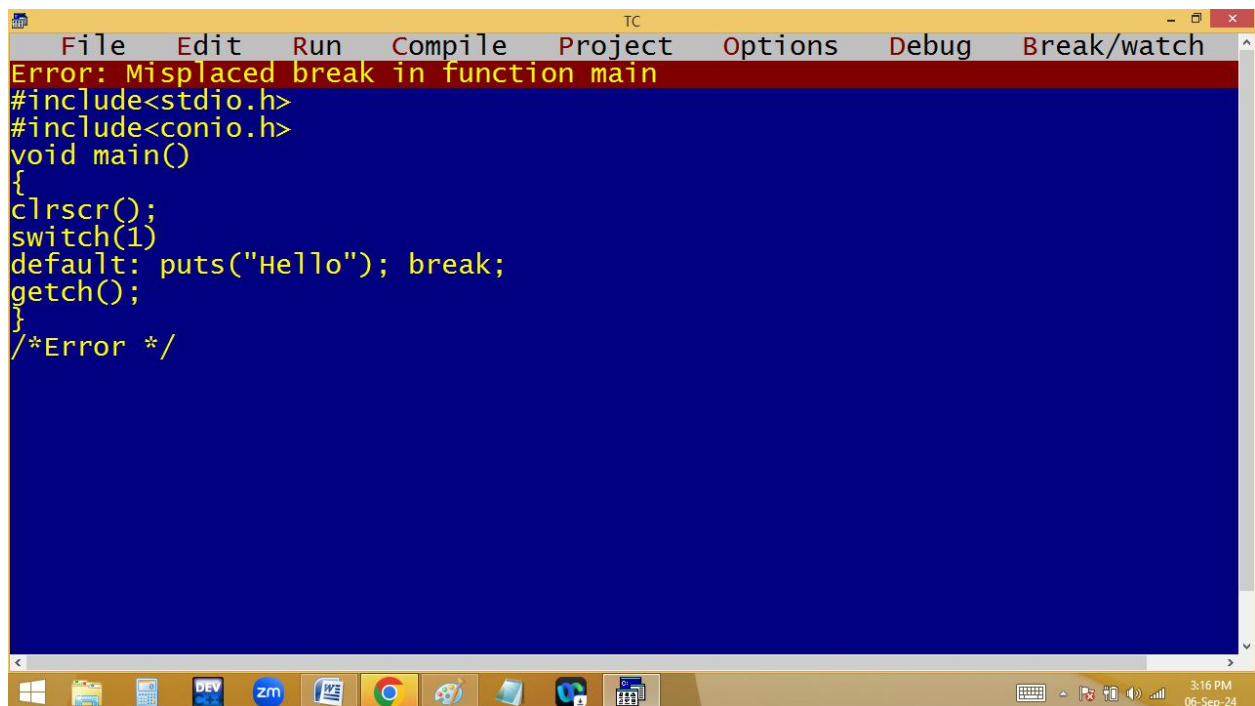
The Windows taskbar at the bottom shows the time as 3:15 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a blue editor window. A red error message bar at the top reads "Error: Case outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
default: puts("Hello");
case 1:  puts("Hi");
getch();
}
/*Error */
```

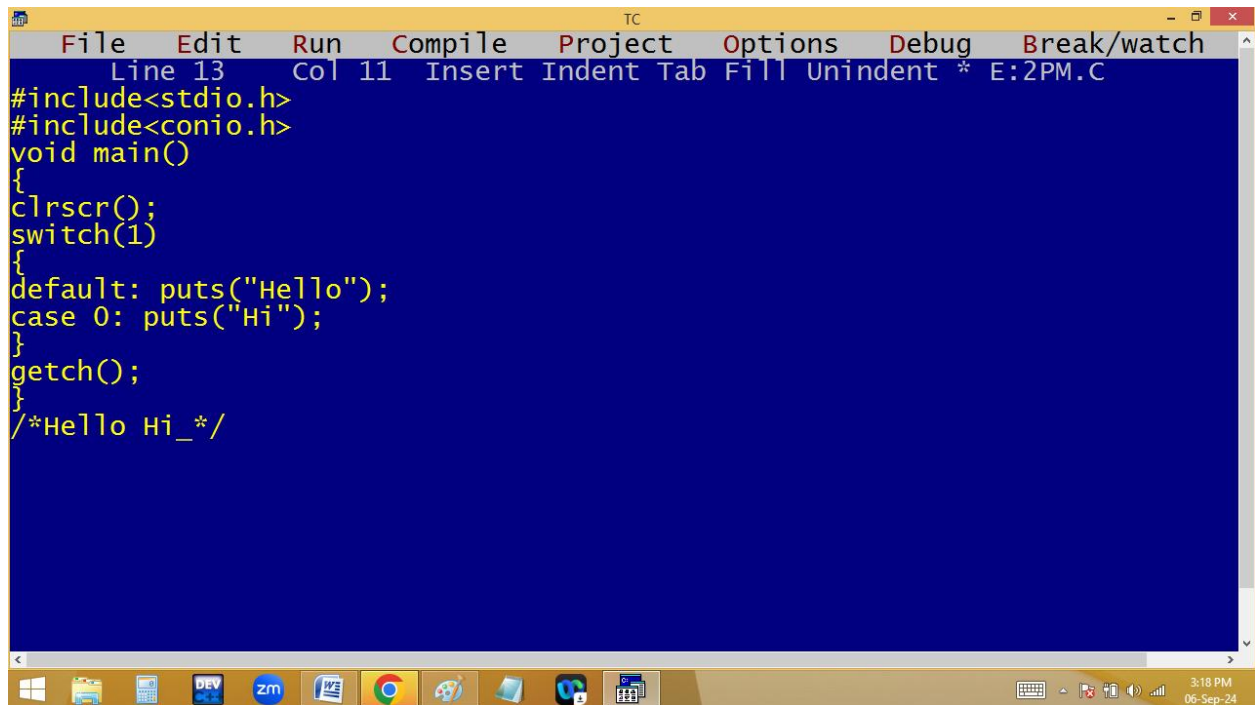
The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:16 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a blue editor window. A red error message bar at the top reads "Error: Misplaced break in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
default: puts("Hello"); break;
getch();
}
/*Error */
```

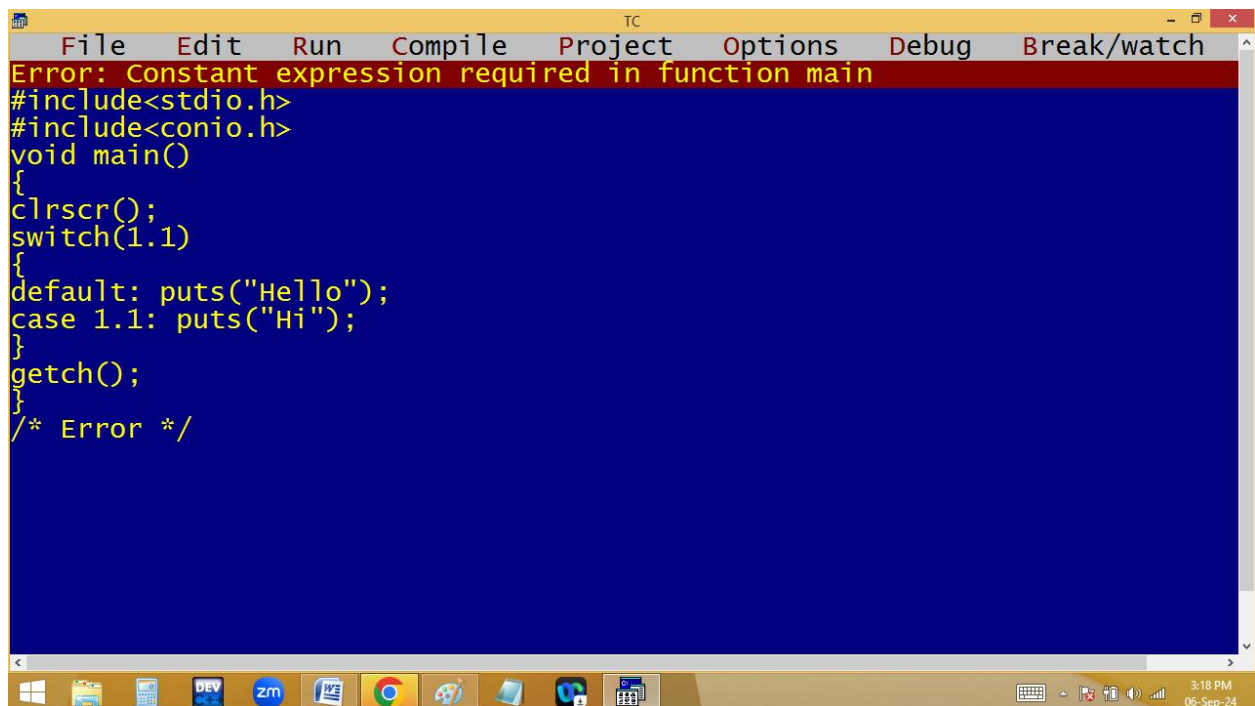
The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:16 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 13 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
default: puts("Hello");
case 0: puts("Hi");
}
getch();
}
/*Hello Hi_*/
```

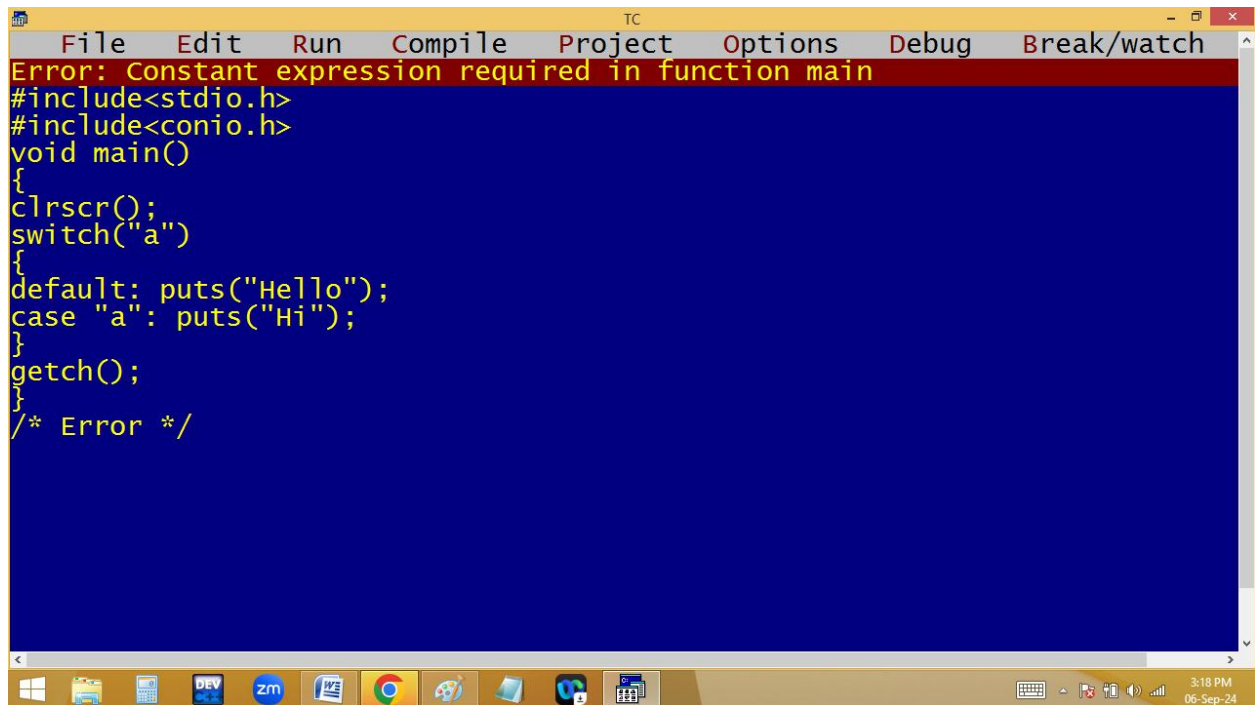
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system clock shows 3:18 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE window after a compilation attempt. A red error message banner is displayed at the top: 'Error: Constant expression required in function main'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1.1)
{
default: puts("Hello");
case 1.1: puts("Hi");
}
getch();
}
/* Error */
```

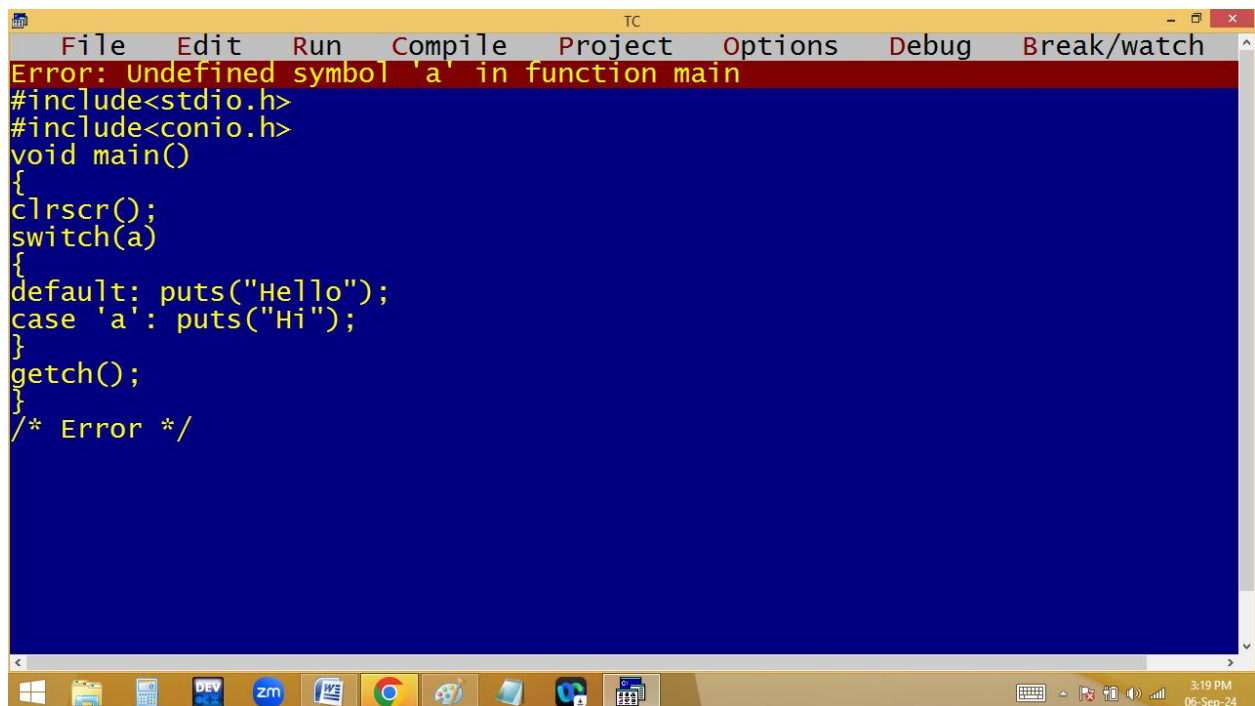
The error message is highlighted in red. The rest of the IDE interface, including the menu bar and Windows taskbar, is identical to the first screenshot.



The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a blue editor window. A red error message is displayed at the top: "Error: Constant expression required in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch("a")
{
default: puts("Hello");
case "a": puts("Hi");
}
getch();
}
/* Error */
```

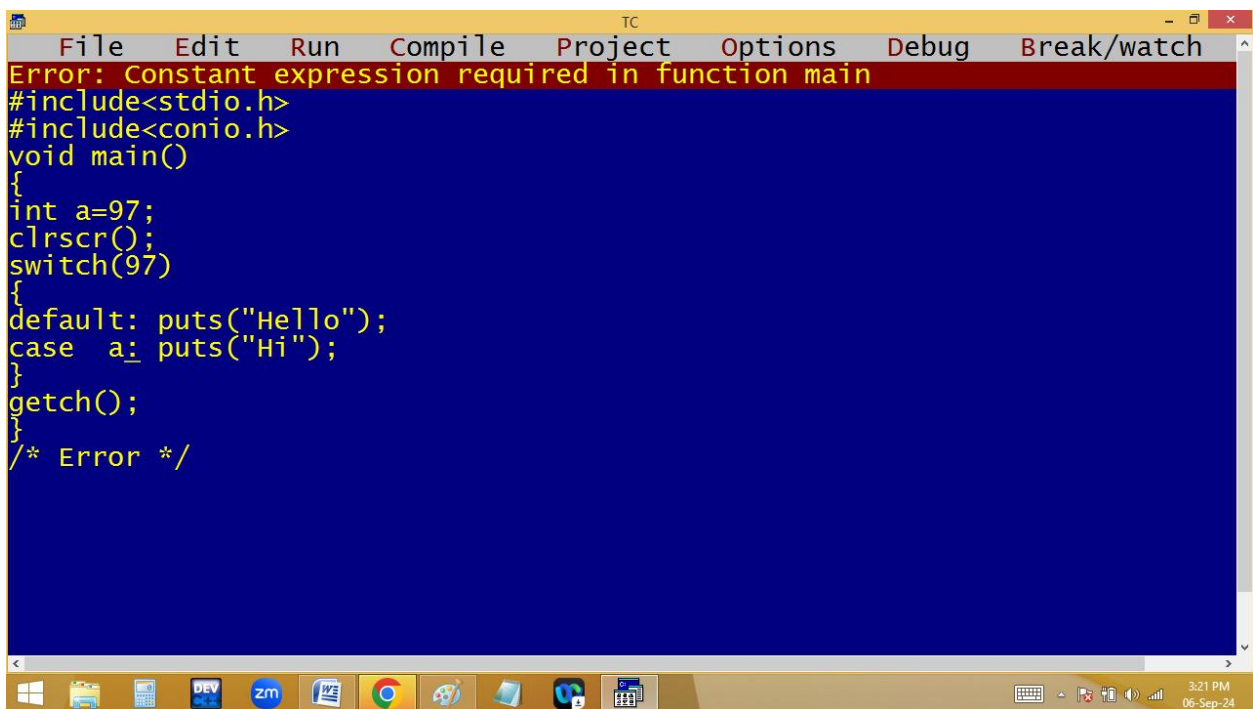
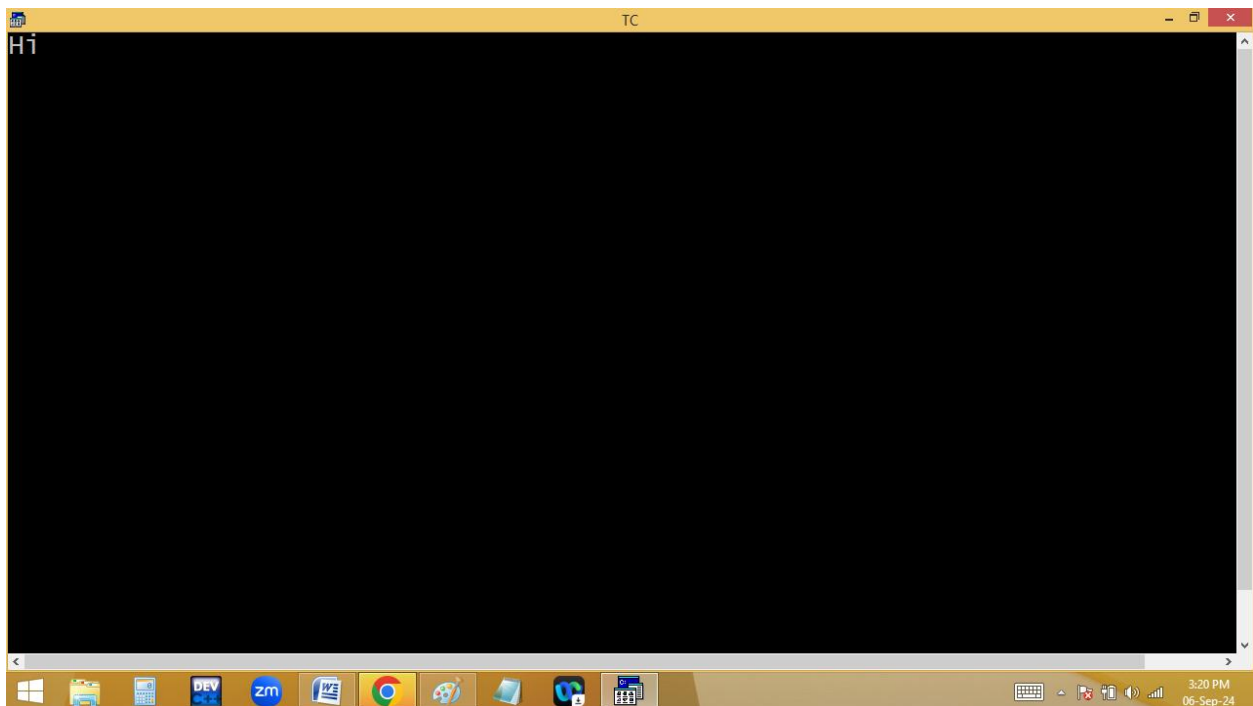
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, WPS, Chrome, and system tray icons for keyboard, volume, and network. The system clock shows 3:18 PM on 06-Sep-24.

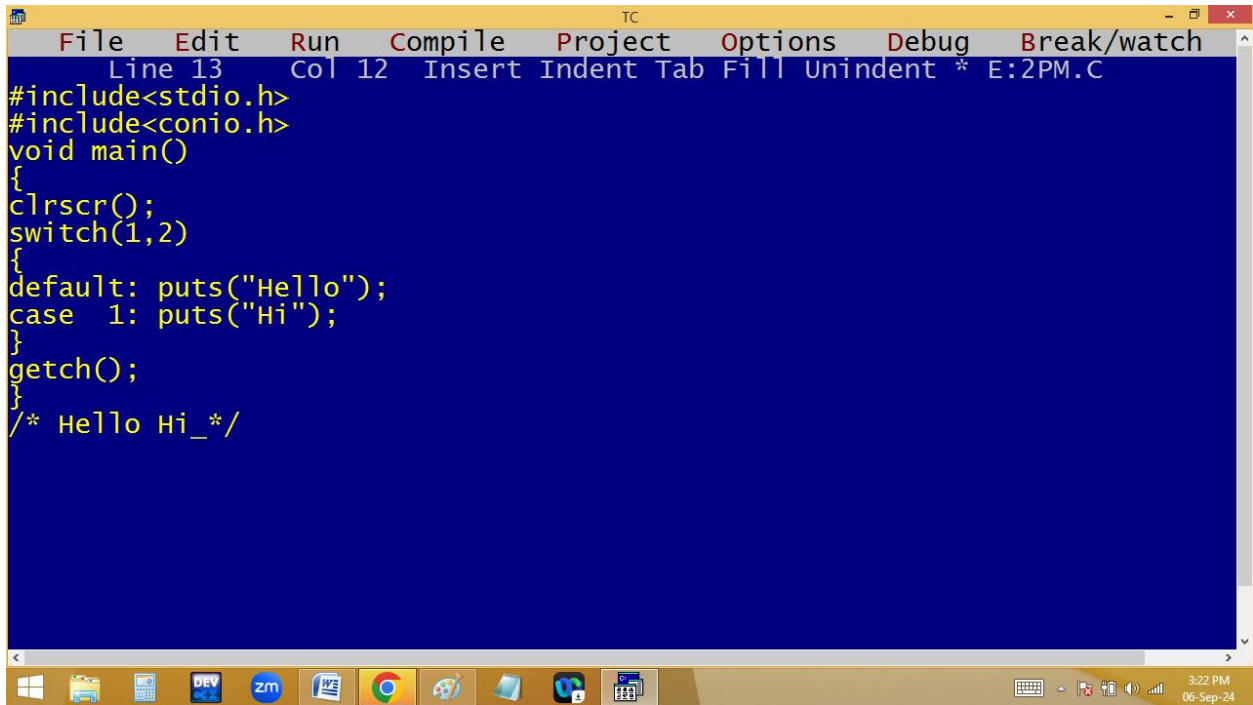


The screenshot shows the Turbo C++ IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a blue editor window. A red error message is displayed at the top: "Error: Undefined symbol 'a' in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(a)
{
default: puts("Hello");
case 'a': puts("Hi");
}
getch();
}
/* Error */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, WPS, Chrome, and system tray icons for keyboard, volume, and network. The system clock shows 3:19 PM on 06-Sep-24.

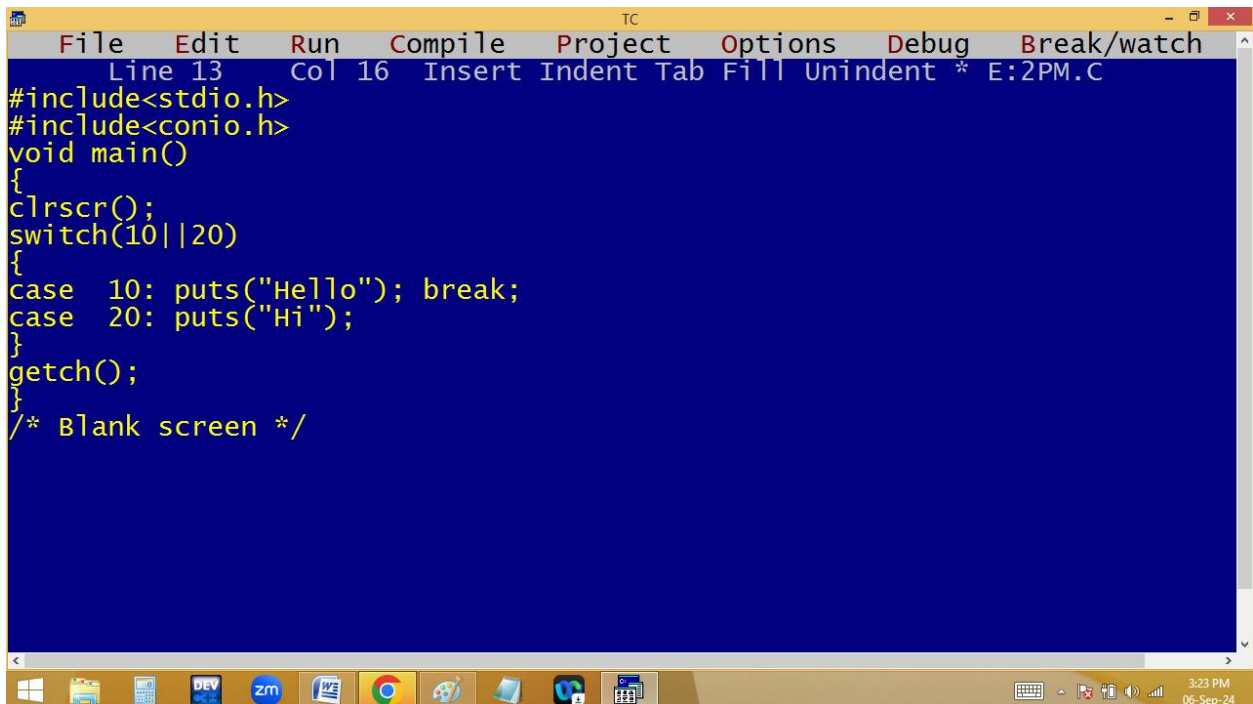




The screenshot shows the Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 13 Col 12 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1,2)
{
default: puts("Hello");
case 1: puts("Hi");
}
getch();
}
/* Hello Hi_*/
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, ZM, Word, Chrome, and others. The system clock in the bottom right corner shows 3:22 PM on 06-Sep-24.



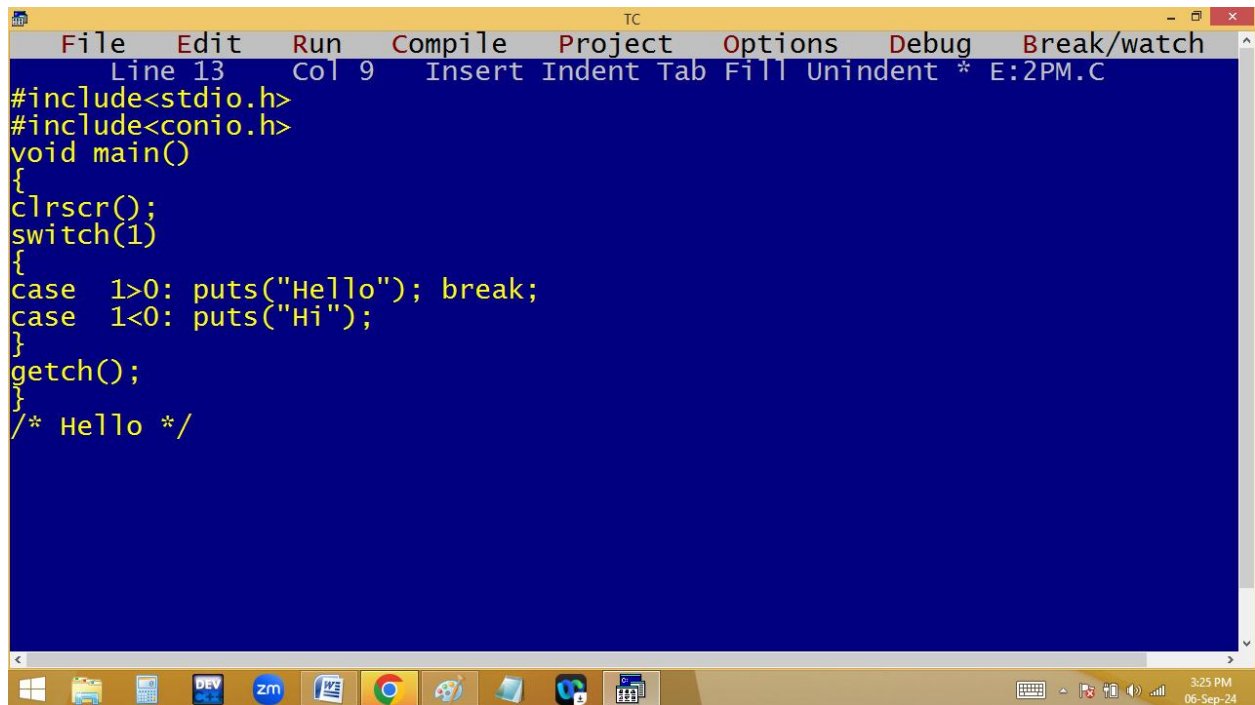
The screenshot shows the Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 13 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(10||20)
{
case 10: puts("Hello"); break;
case 20: puts("Hi");
}
getch();
}
/* Blank screen */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, ZM, Word, Chrome, and others. The system clock in the bottom right corner shows 3:23 PM on 06-Sep-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Duplicate case in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(10||20)
{
case 10||20: puts("Hello"); break;
case 20||10: puts("Hi");
}
getch();
}
/* Error */
```

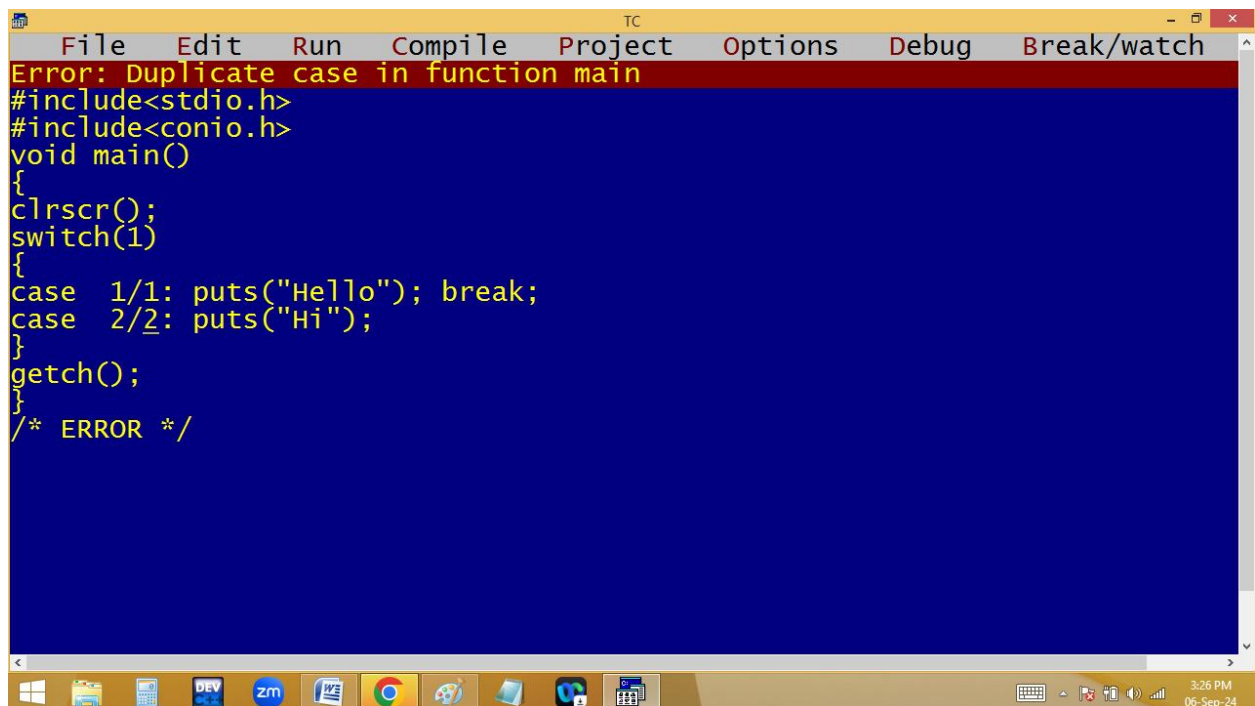
```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(3)
{
case >0: puts("Hello"); break;
case <0: puts("Hi");
}
getch();
}
/* Error */
```



The screenshot shows the Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 13 Col 9' and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1>0: puts("Hello"); break;
case 1<0: puts("Hi");
}
getch();
}
/* Hello */
```

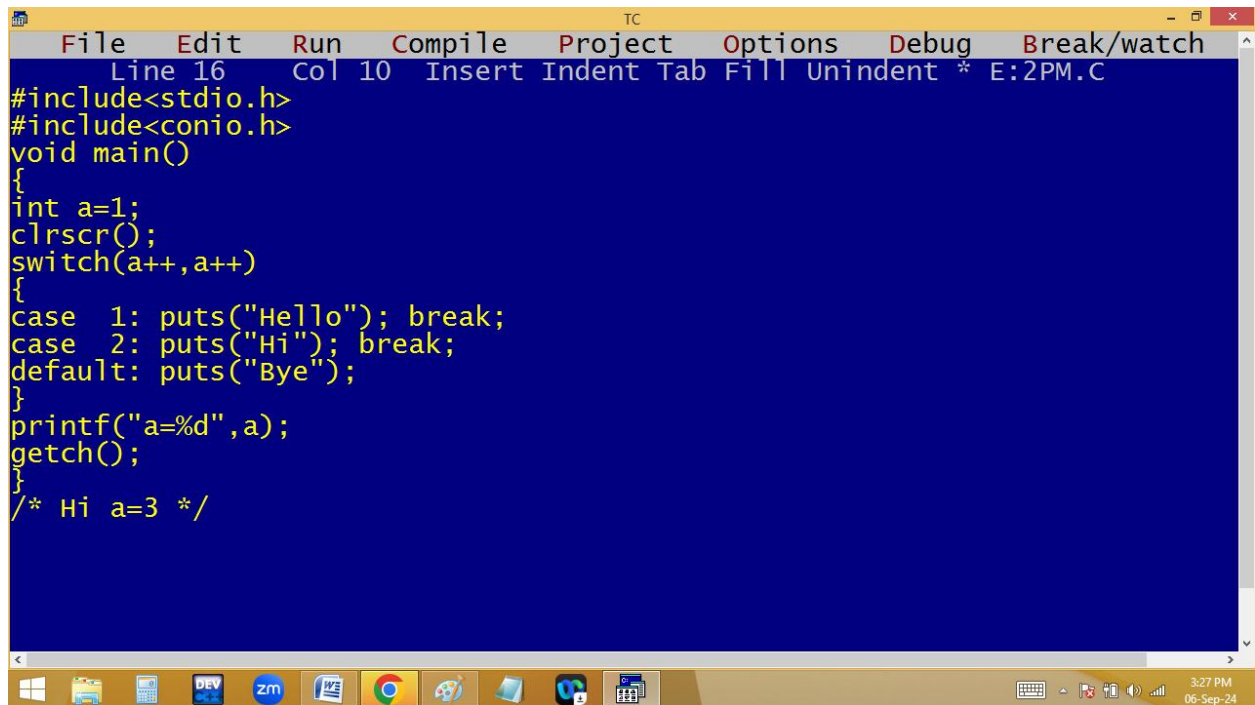
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and others. The system clock shows 3:25 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE window with a compilation error. A red error message banner at the top reads: "Error: Duplicate case in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1/1: puts("Hello"); break;
case 2/2: puts("Hi");
}
getch();
}
/* ERROR */
```

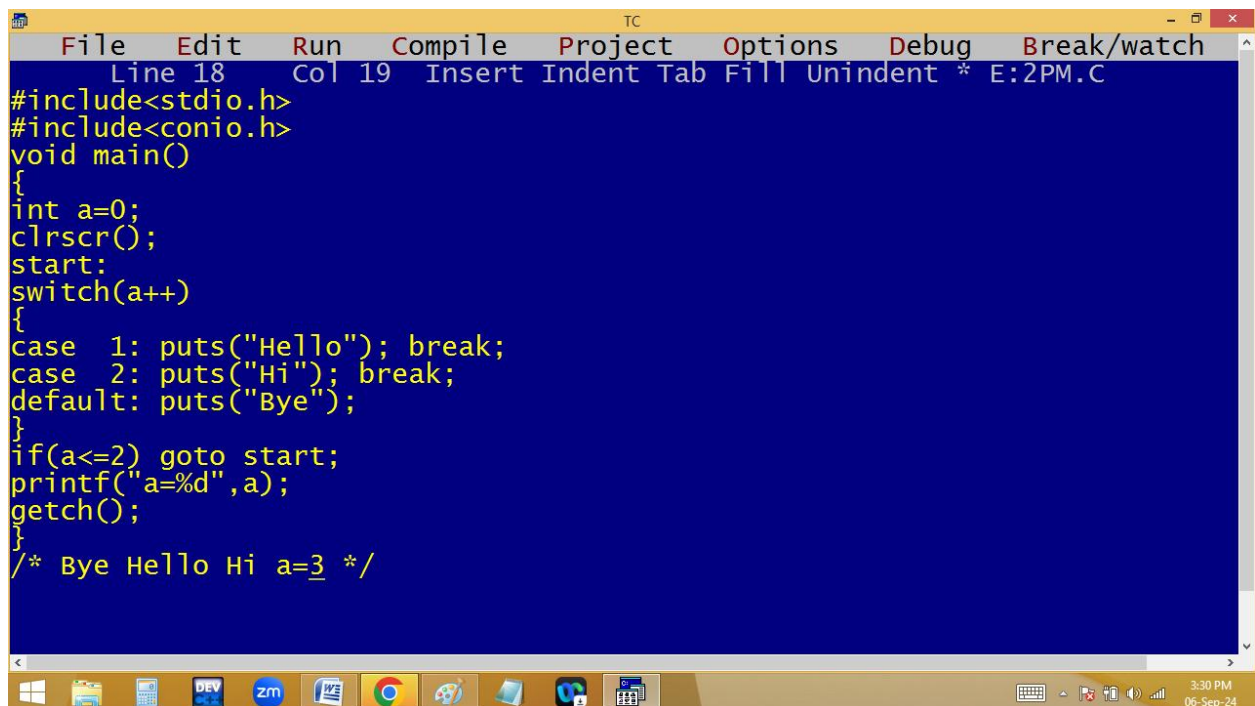
The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock (3:26 PM on 06-Sep-24).



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 16, Col 10, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=1;
    clrscr();
    switch(a++,a++)
    {
        case 1: puts("Hello"); break;
        case 2: puts("Hi"); break;
        default: puts("Bye");
    }
    printf("a=%d",a);
    getch();
}
/* Hi a=3 */
```

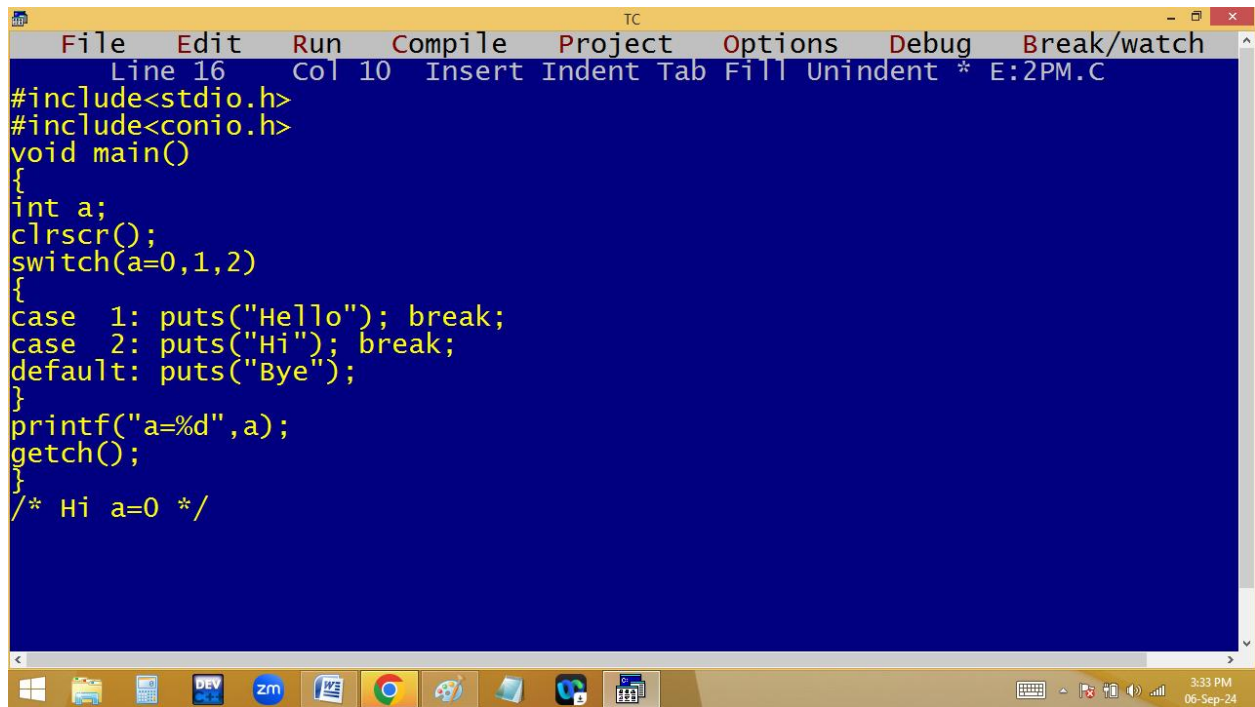
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system clock indicates 3:27 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 18, Col 19, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=0;
    clrscr();
    start:
    switch(a++)
    {
        case 1: puts("Hello"); break;
        case 2: puts("Hi"); break;
        default: puts("Bye");
    }
    if(a<=2) goto start;
    printf("a=%d",a);
    getch();
}
/* Bye Hello Hi a=3 */
```

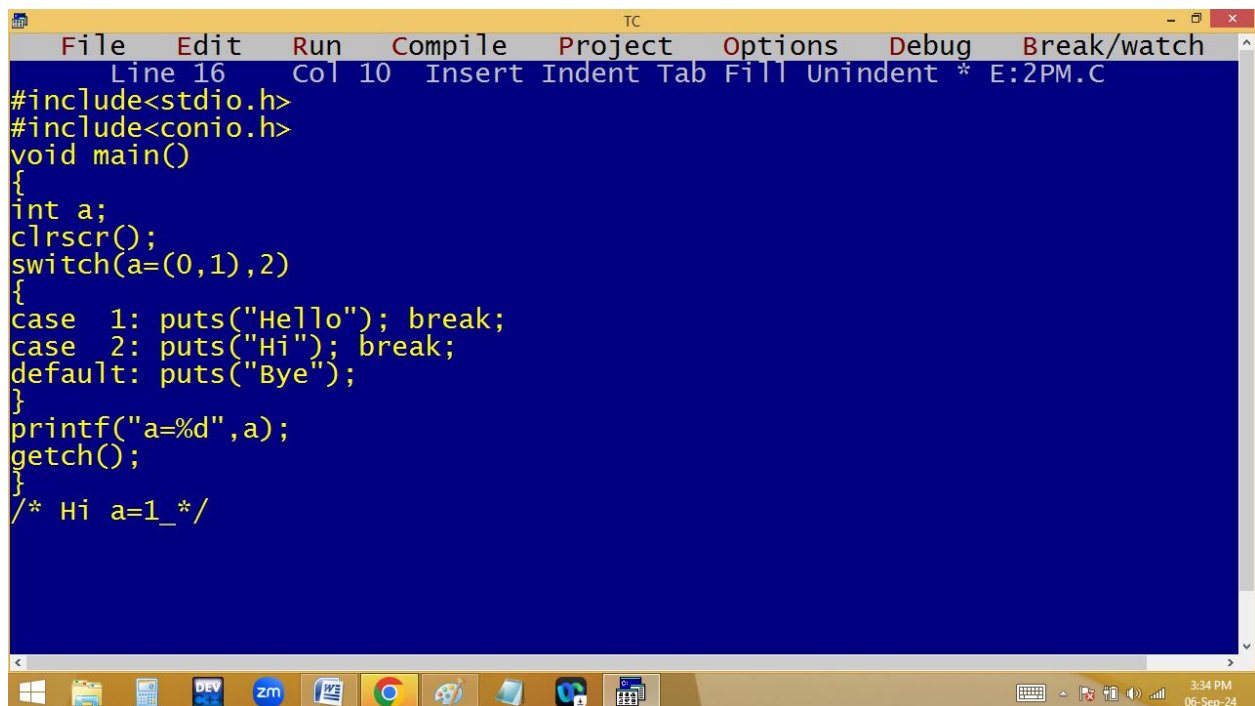
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system clock indicates 3:30 PM on 06-Sep-24.



The screenshot shows the Turbo C++ IDE with the file E:2PM.C open. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
switch(a=0,1,2)
{
case 1: puts("Hello"); break;
case 2: puts("Hi"); break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}
/* Hi a=0 */
```

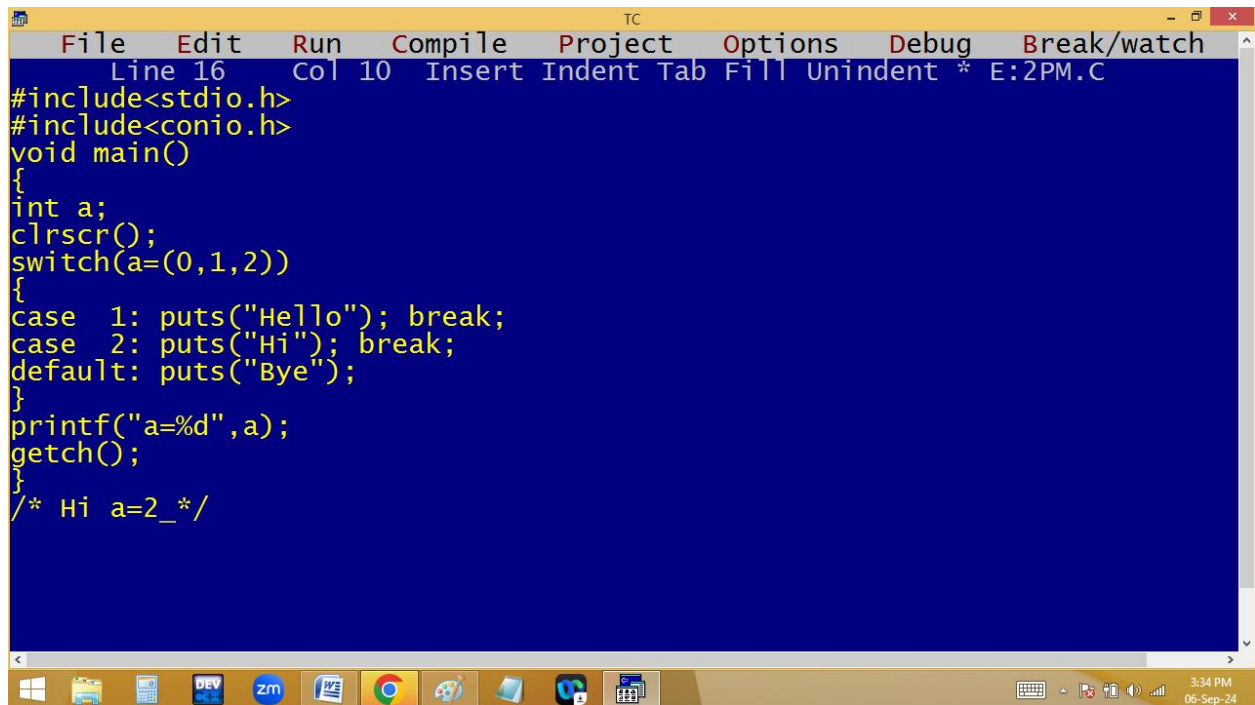
The taskbar at the bottom shows the Windows Start button, taskbar icons for File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and other applications. The system clock shows 3:33 PM on 06-Sep-24.



The screenshot shows the Turbo C++ IDE with the file E:2PM.C open. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
switch(a=(0,1),2)
{
case 1: puts("Hello"); break;
case 2: puts("Hi"); break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}
/* Hi a=1_*/
```

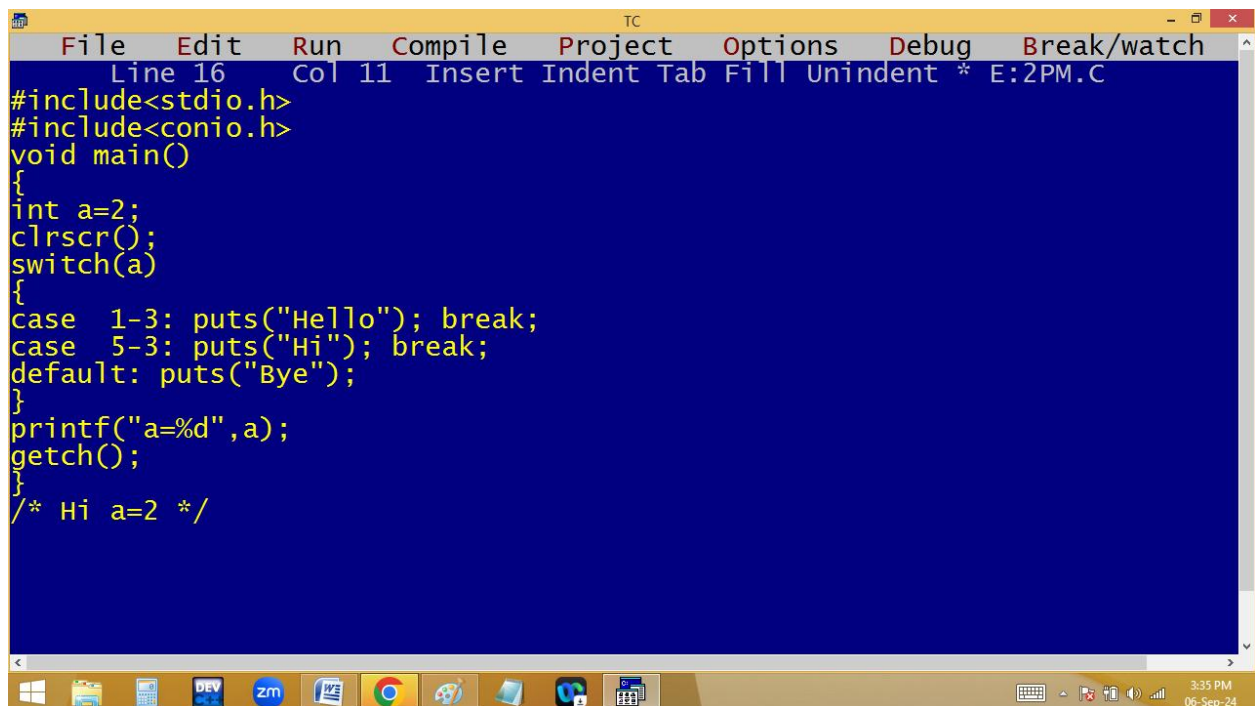
The taskbar at the bottom shows the Windows Start button, taskbar icons for File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and other applications. The system clock shows 3:34 PM on 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 16, Col 10, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a;
    clrscr();
    switch(a=(0,1,2))
    {
        case 1: puts("Hello"); break;
        case 2: puts("Hi"); break;
        default: puts("Bye");
    }
    printf("a=%d",a);
    getch();
}
/* Hi a=2_*/
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system clock indicates 3:34 PM on 06-Sep-24.

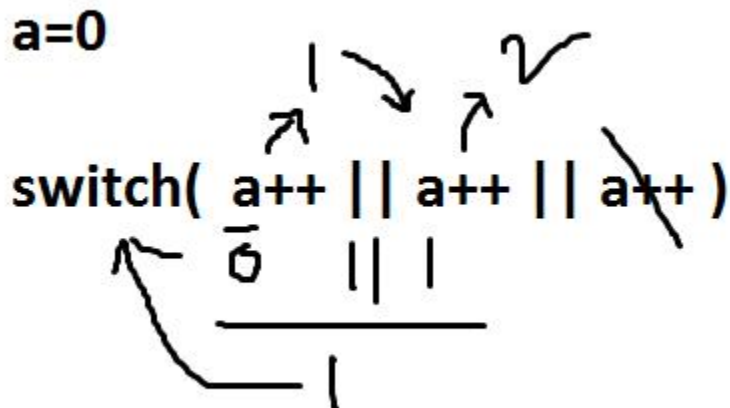


The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 16, Col 11, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=2;
    clrscr();
    switch(a)
    {
        case 1-3: puts("Hello"); break;
        case 5-3: puts("Hi"); break;
        default: puts("Bye");
    }
    printf("a=%d",a);
    getch();
}
/* Hi a=2 */
```

The Windows taskbar at the bottom shows various icons including Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, and others. The system clock indicates 3:35 PM on 06-Sep-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 26 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a++ || a++ || a++)
{
case 1: puts("Hello"); break;
case 2: puts("Hi"); break;
case 3: puts("welcome"); break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}
/* Hello a=2 */
```



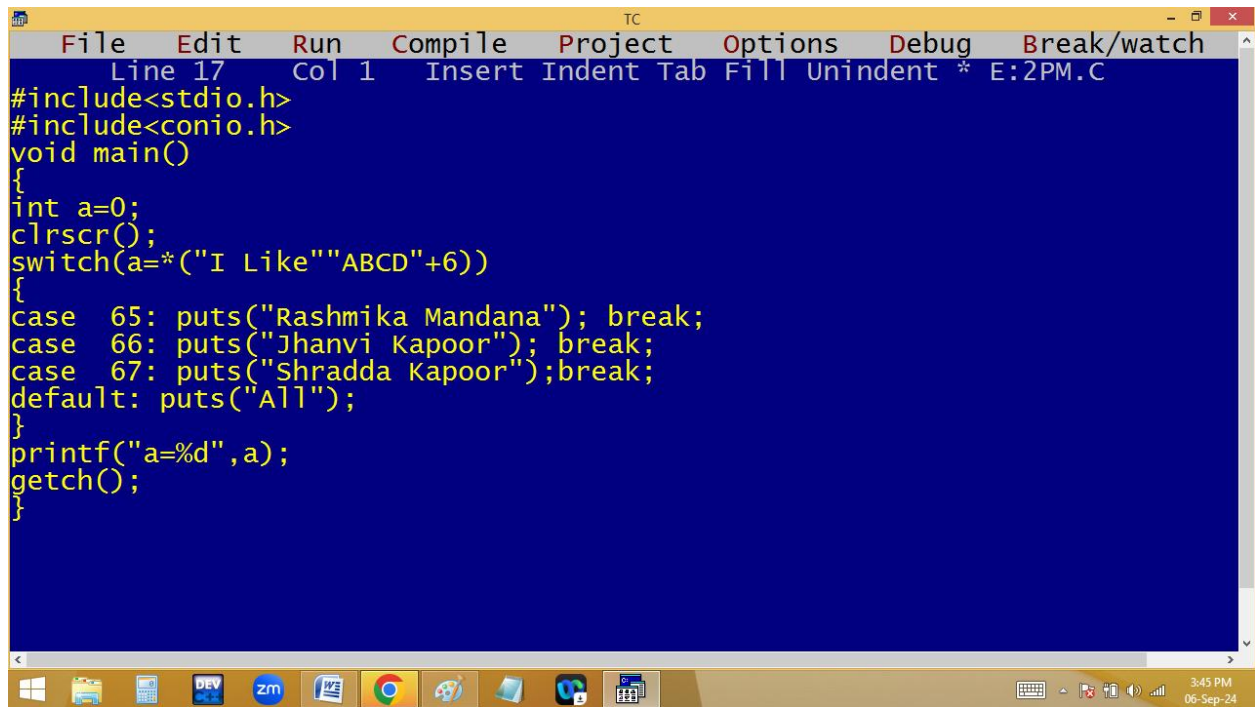
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=printf("I\t"), printf("Am\t"))
{
case 1: puts("Bharathiyudu"); break;
case 2: puts("Indian"); break;
case 3: puts("Hindustani");break;
default: puts("Human Being");
}
printf("a=%d",a);
getch();
}
/* I am Hindustani a=2_*/
```

✓ ✓

switch(a = p("I\t"), p("Am\t"))

a = 2, 3

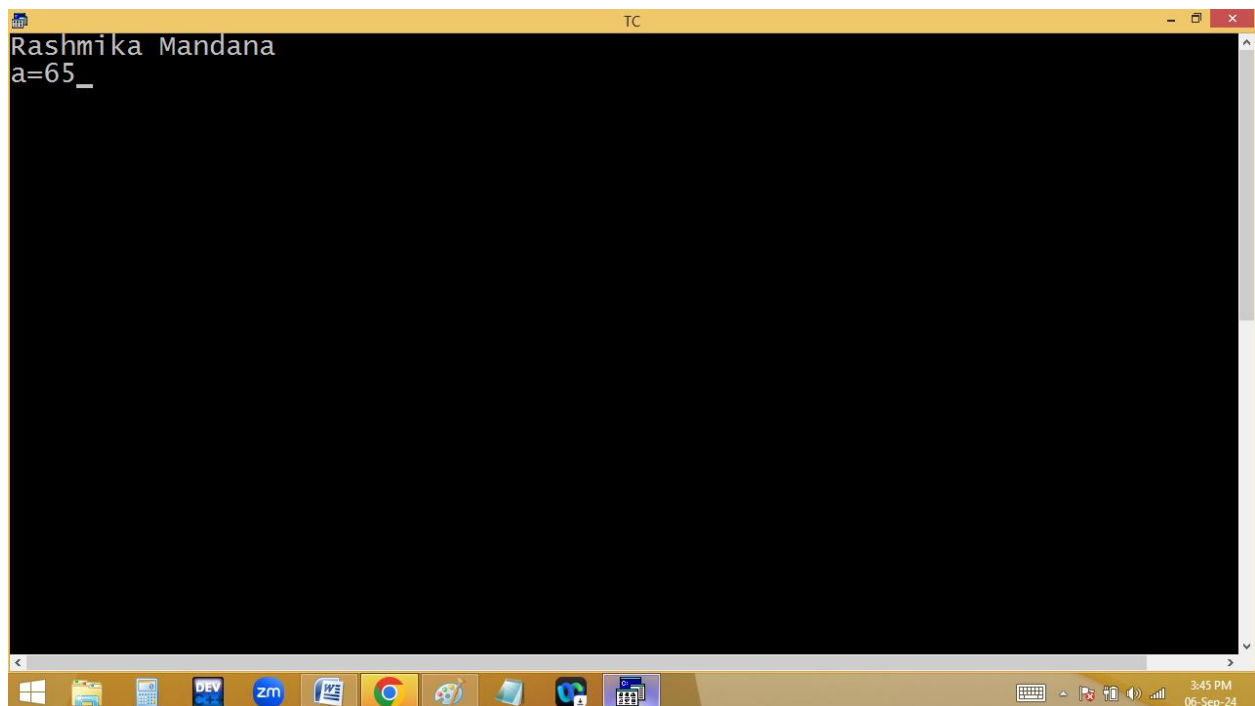




The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 17, Col 1, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=0;
    clrscr();
    switch(a*("I Like""ABCD"+6))
    {
        case 65: puts("Rashmika Mandana"); break;
        case 66: puts("Jhanvi Kapoor"); break;
        case 67: puts("Shradda Kapoor");break;
        default: puts("All");
    }
    printf("a=%d",a);
    getch();
}
```

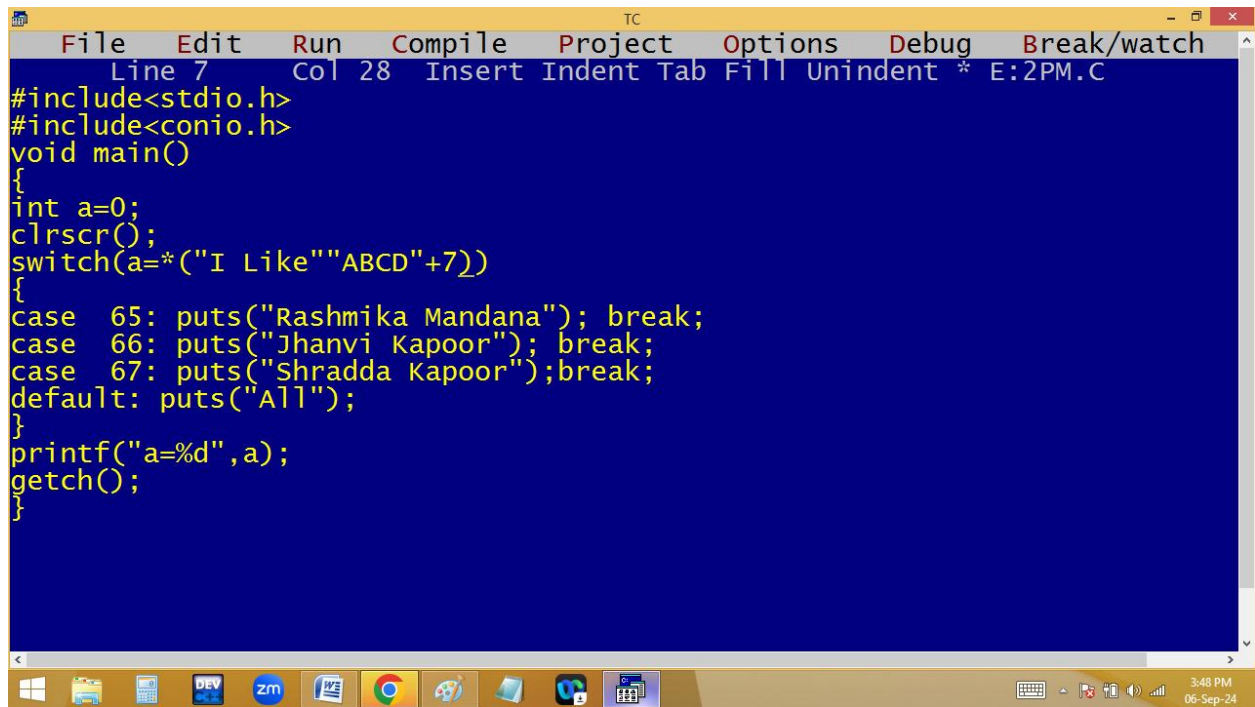
The Windows taskbar at the bottom shows icons for Windows, File Explorer, Calculator, DEV C++, ZOOM, Word, Chrome, Paint, and a folder icon. The system tray on the right shows the time as 3:45 PM and the date as 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar. The output window displays the results of the program execution:

```
Rashmika Mandana
a=65_
```

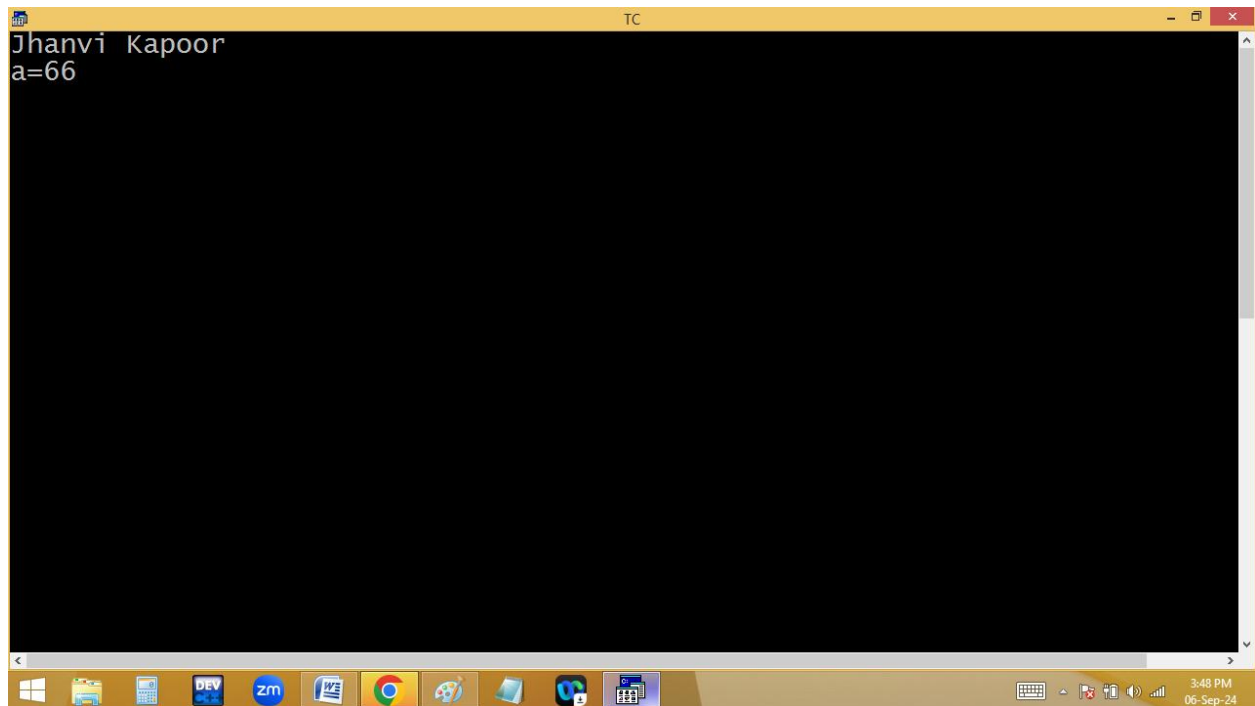
The Windows taskbar and system tray are identical to the first screenshot, showing the time as 3:45 PM and the date as 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 7, Col 28, Insert, Indent, Tab, Fill, Unindent, *, E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=0;
    clrscr();
    switch(a=*"I Like""ABCD"+7)
    {
        case 65: puts("Rashmika Mandana"); break;
        case 66: puts("Jhanvi Kapoor"); break;
        case 67: puts("Shradda Kapoor");break;
        default: puts("All");
    }
    printf("a=%d",a);
    getch();
}
```

The Windows taskbar at the bottom shows icons for Windows, File Explorer, Calculator, DEV C++, Zm, Word, Chrome, Paint, and a folder icon. The system tray on the right shows the keyboard icon, network status, volume, and the date/time: 3:48 PM, 06-Sep-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar as the first image. The output window displays the results of the program's execution:

```
Jhanvi Kapoor
a=66
```

The Windows taskbar and system tray are identical to the first image, showing the same icons and date/time (3:48 PM, 06-Sep-24).

